1. Metals

1.1. Properties of Metals

Metals

Physical Properties

1. Good conductors of heat and electricity.
2. Lustrous
3. Malleable and Ductile
4. High densities and High Melting Points
5. Solids at room temperature (except mercury is liquid)

Chemical Properties

1. Metals + Acids → Salt + Hydrogen
2. Metal + Oxygen → Metal Oxide
3. Metal + Water → Metal Hydroxide + Hydrogen
4. Metal + Steam → Metal Oxide + Hydrogen

Properties of Non Metals

Physical Properties

1. Poor thermal conductor
2. Poor conductor of electricity (Except graphite)
3. Low Density and Low Melting Point
4. Brittle and not malleable
5. Solids and gases at room temperature (bromine is liquid)

1.2. Uses of Metals

1. Aluminium
   - Manufacture of Aircraft/Cars (Low density)
   - Food Containers (Resistant to corrosion)
   - Overhead electrical cable (Good conductor of electricity/ductile)
2. Zinc
   - Galvanizes Iron = coats it to stop it from rusting (protective coating)
   - Alloys – brass/bronze
   - Batteries
   - Sacrificial Protection
3. Copper
   - Electrical Wiring (Good conductor of electricity/Ductile)

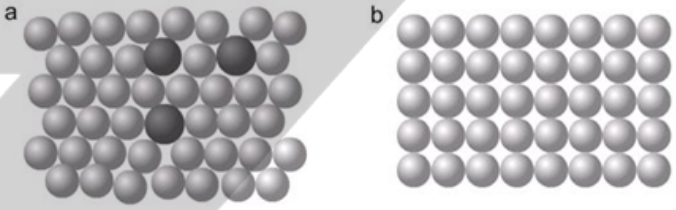
1.3. Alloys and their Properties

Alloy: a mixture of two or more metals or a metal and non-metal

- Alloys are useful because they are harder and stronger/useful than pure metal; they can also resist rusting with air and water.

Alloy	Made from	Special Properties	Uses
Brass	Copper and zinc	Stronger and more resistant to corrosion/rusting	Electrical fittings, car radiators
Stainless steel	Iron, carbon, chromium and nickel	resistant to corrosion/rusting	Kitchen sinks, cutlery, surgical instruments

- Due to the irregularity in atom sizes and structure, metal alloys, which stop layers from sliding over each other, are stronger.
- Alloys contain metals cations of different sizes, the large cations block the layer of small metal cations.
- This is what the structure of an alloy (a) looks like compared to a pure metal (b).



1.4. Reactivity Series

NOTE: Aluminum - despite its high placement in the reactivity series, it is seemingly unreactive due to its protective aluminium oxide layer.

General Reactivity Series (descending order of reactivity)	The Chemical Reactivity of Metals
K - Potassium Na - Sodium Ca – Calcium Mg – Magnesium Al – Aluminum C – Carbon Zn – Zinc Fe – Iron Pb – Lead H – Hydrogen Cu – Copper Ag – Silver Au – Gold	Water: 1. metal + cold water → metal hydroxide + hydrogen 2. metal + steam → metal oxide + hydrogen
Everything above hydrogen can displace hydrogen from its acid, and hydrogen cannot reduce its oxides.	Oxygen: metal + oxygen → metal oxide
Metals above carbon, their oxides cannot be reduced by carbon	Dilute acids: In a metal and acid reaction, the hydrogen atom in the acid is replaced by the metal atom to form a product of salt and hydrogen

Because they are too unreactive, Copper, Silver and Gold cannot react with Dilute Hydrochloric acid.

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Displacement Reactions

In these reactions, metals compete for oxygen or anions

- Oxidisation is the loss of electrons
- Reduction is the gain of electrons
- The more reactive metal will displace the less reactive metal from oxygen or an anion.
- If the more reactive metal has oxygen or an anion, no reaction occurs
- The bigger the difference in reactivity between the two metals, the faster the reaction

Example: $Mg + O_2 \rightarrow 2MgO$
Here magnesium is oxidised while the oxygen atom is reduced

1.5. Corrosion of Metals

Rusting: the corrosion of iron and steel to form rust (**hydrated iron (III) oxide**) via oxidation

The conditions for rusting to occur:

1. Water
2. Air containing oxygen

The Equation for Rusting of Iron

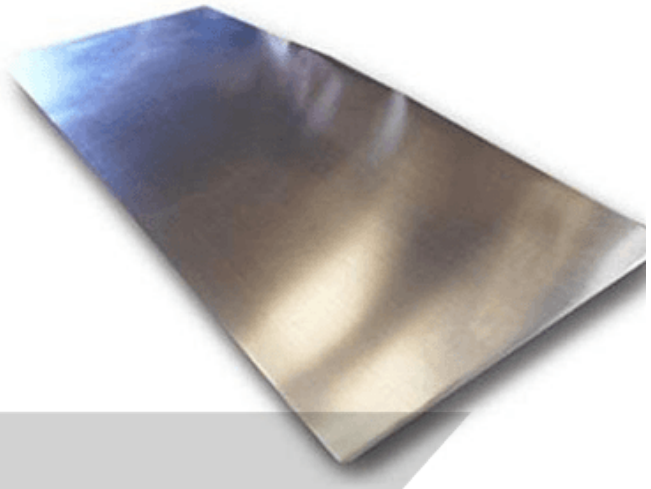
Iron + Water + Oxygen → Hydrated Iron (III) Oxide
$4Fe\ (s) + 2H_2O\ (l) + 3O_2\ (g) \rightarrow 2Fe_2O_3{\cdot}H_2O\ (s)$

Two Ways of Preventing Rust

1. **Barrier Method**
   - Coating a material to prevent the iron/steel from being in contact with water and oxygen (Painting, Greasing & Plastic Coating)
2. **Sacrificial Method**
   - When more reactive metals corrode to less reactive metals, they lose electrons in preference to iron.

Galvanisation of Iron

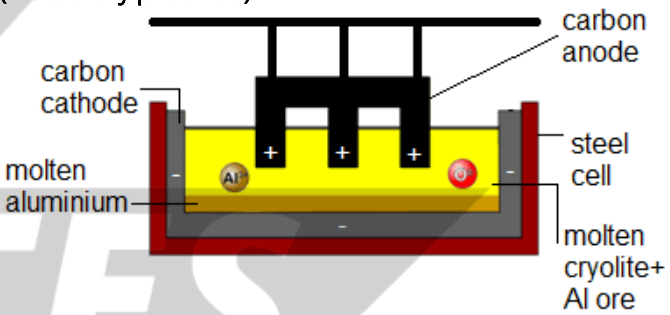
It is the **protection of iron and steel objects** by coating them with a **layer of zinc.**



The barrier method is due to the zinc layer preventing exposure to air and water and sacrificial protection due to zinc being more reactive than iron, corroding in preference to iron.

1.6. Extraction of Aluminium

- The main ore of Aluminium Oxide is bauxite
- Aluminum (III) oxide (alumina) is dissolved in molten **cryolite** (Na_3AlF_6) **– this mixture has a lower melting point (industrially preferred)**



During electrolysis, aluminium ($Al^{3+} + 3e- \rightarrow Al$) is produced at the carbon cathode, and oxygen ($2O^{2-} \rightarrow O_2 + 4e-$) at the carbon anode.

Molten Cryolite

- High temperatures are needed to melt, which is expensive. Therefore, molten cryolite **lowers the melting point/operating temperature and increases conductivity.**
- Due to the high temperature, the oxygen **reacts with the carbon** in the graphite anode to form CO_2, so the anode must **be periodically replaced.**

1.7. Extraction of Metals

- Process of separating a particular metal from its compound; metal ore

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- Ore is more difficult to decompose from gold to potassium (expensive)

Metal	Extraction Method
K - Potassium Na - Sodium Ca – Calcium Mg – Magnesium Al – Aluminum	Reduction via electrolysis
Carbon	
Zn – Zinc Fe – Iron Pb – Lead	Reducing by Carbon
Hydrogen	
Cu – Copper Ag – Silver Au – Gold	Occur naturally

1.8. Extraction of Iron

- The Ore of Iron is hematite (Fe_2O_3) + Impurities
- Burning of coke (Carbon) to **provide heat and produce carbon dioxide**

1. Ore is crushed and mixed with carbon and limestone ($CaCO_3$) and transferred into the blast furnace

$C + O_2 \rightarrow CO_2$ (exothermic)
$CaCO_3 \rightarrow CaO + CO_2$ (thermal decomposition)
$CO_2 + C \rightarrow 2CO$

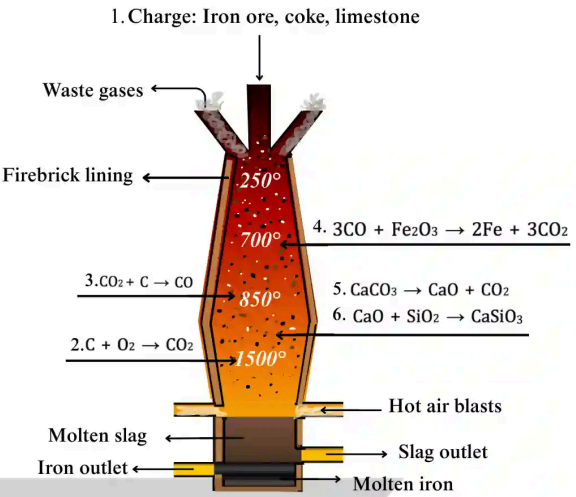
2. Carbon monoxide reduces Iron(III) oxide from the ore to iron

   $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

3. The Calcium oxide reacts with impurities like silica to form slag (a waste product) - Neutralization reaction

   $CaO + SiO_2 \rightarrow CaSiO_3$

4. Uses of slag include making roads and cement



The remaining waste gases that leave the blast furnace are **Carbon Dioxide, Carbon Monoxide, and Nitrogen.**

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